

C672

General Chemistry I with Lab

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A teacher who keeps a periodic table poster from 1995 on the wall asks for the ground-state electron configuration of a neutral sulfur atom (atomic number 16). Sulfur is mined in large amounts near volcanoes. Which configuration is correct?

- A. $1s^2 2s^2 2p^6 3s^2 3p^6$
- B. $1s^2 2s^2 2p^6 3s^2 3d^4$
- C. $1s^2 2s^2 2p^6 3p^6$
- D. $1s^2 2s^2 2p^6 3s^2 3p^4$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

An element's ground-state configuration is $[\text{Ne}] 3s^2 3p^3$. Which element is it?

- A. Neon
- B. Nitrogen
- C. Phosphorus
- D. Sulfur

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A lab group in a school that just renovated its science wing compares four period 3 elements: Na, Mg, P, and Cl. Their teacher mentions that phosphorus is used in fertilizer. Which has the highest first ionization energy?

- A. Mg, because its 3s subshell is full
- B. P, because its half-filled 3p subshell is extra stable
- C. Cl
- D. Na

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A student writing a lab report on a corrosion-resistant coating needs the formula of aluminum oxide. The coating was tested for 30 days. Aluminum forms Al^{3+} and oxygen forms O^{2-} . What is the formula?

- A. Al_3O_2
- B. AlO_3
- C. Al_2O_3
- D. AlO

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

What is the name of Fe_2O_3 ?

- A. Iron(II) oxide
- B. Iron oxide
- C. Diiron trioxide
- D. Iron(III) oxide

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A teacher who orders model kits every August asks students to build ammonia, NH_3 . The central nitrogen has three bonds and one lone pair. The kit uses blue balls for nitrogen. What is the molecular geometry?

- A. Bent
- B. Trigonal pyramidal
- C. Trigonal planar
- D. Tetrahedral

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

Propane from a 20-pound grill tank burns: $\text{C}_3\text{H}_8 + __\text{O}_2 \rightarrow 3\text{CO}_2 + 4\text{H}_2\text{O}$. The grill was bought on sale. What coefficient balances O_2 ?

- A. 7
- B. 5
- C. Change 3CO_2 to 3CO so fewer O atoms are needed.
- D. 10

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

In a demonstration that takes place behind a safety shield, 4.0 g of H_2 reacts with excess O_2 :
 $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$. The room holds 24 students. What mass of water forms? ($\text{H}_2 = 2.0 \text{ g/mol}$,
 $\text{H}_2\text{O} = 18.0 \text{ g/mol}$)

- A. 18 g
- B. 36 g
- C. 72 g
- D. 4.0 g

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

For $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$, how many grams of O_2 are needed to react completely with 4.0 g of H_2 ?
($\text{H}_2 = 2.0 \text{ g/mol}$, $\text{O}_2 = 32.0 \text{ g/mol}$)

- A. 32 g
- B. 64 g
- C. 16 g
- D. 4.0 g

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

Students heat ice from -20°C using a hot plate with a digital display. At 0°C the thermometer holds steady for several minutes while heating continues. The lab has two windows. What is the added energy doing during that time?

- A. Escaping to the room, since no energy enters the ice
- B. Raising the temperature too slowly for the thermometer
- C. Breaking the O-H bonds inside each water molecule
- D. Breaking attractions between water molecules as the ice melts

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

Energy is added steadily to a pure substance, but its temperature does not change for a while. What is happening?

- A.** It is changing phase.
- B.** Its specific heat is zero.
- C.** It is undergoing a chemical reaction.
- D.** It is losing energy to the surroundings.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A teacher preparing 1 M sulfuric acid for a lab of 30 students has a bottle of concentrated acid, bought from a supplier the school has used for years, and a flask of distilled water. What is the safe procedure?

- A.** Slowly add the acid to the water with goggles on, stirring as it warms.
- B.** Mix them in any order, since the final concentration is the same.
- C.** Pour the water into the acid to dilute it quickly.
- D.** Ask students to do the dilution so they practice the skill.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

A teacher ~~who keeps a periodic table poster from 1995 on the wall~~ asks for the ground-state electron configuration of a neutral sulfur atom (atomic number 16). ~~Sulfur is mined in large amounts near volcanoes.~~ Which configuration is correct?

Target word: configuration for $Z = 16$ **Decides:** 16 electrons in order.

A	Wrong stem number	18 electrons: this is the sulfide ion (S^{2-}) or argon.
B	Confusion set	3p fills before 3d.
C	Dropped element	Skips the 3s subshell.
D	Correct	$2 + 2 + 6 + 2 + 4 = 16$.

2. Answer C Concept 1 • transfer

An element's ground-state configuration is $[Ne] 3s^2 3p^3$. Which element is it?

Target word: element from configuration **Decides:** $10 + 5 = 15$ electrons.

A	Wrong stem number	That is only the core.
B	Confusion set	Same valence pattern ($s^2 p^3$) but one period up.
C	Correct	$Z = 15$.
D	Dropped element	Miscounts by one electron.

3. Answer C Concept 2

A lab group ~~in a school that just renovated its science wing~~ compares four period 3 elements: Na, Mg, P, and Cl. ~~Their teacher mentions that phosphorus is used in fertilizer.~~ Which has the highest first ionization energy?

Target word: highest ionization energy **Decides:** Ionization energy rises across a period.

A	Confusion set	Explains $Mg > Al$, not $Mg > Cl$.
B	Adjacent category	That exception puts P above S, not above Cl.
C	Correct	Farthest right; highest effective nuclear charge in the list.
D	Inverted operation	Lowest: one loosely held valence electron.

4. Answer C Concept 3

A student writing a lab report ~~on a corrosion-resistant coating~~ needs the formula of aluminum oxide. ~~The coating was tested for 30 days.~~ Aluminum forms Al^{3+} and oxygen forms O^{2-} . What is the formula?

Target word: ionic formula **Decides:** $2(+3) + 3(-2) = 0$.

A	Inverted operation	Swaps the subscripts: +9 and -4 do not balance.
B	Wrong stem number	Uses the aluminum charge as the oxygen subscript only.
C	Correct	+6 and -6 balance.
D	Dropped element	One of each ion leaves a +1 charge.

5. Answer D Concept 3 · transfer

What is the name of Fe_2O_3 ?

Target word: name from formula **Decides:** Three $\text{O}^{2-} = -6$, shared by two Fe, so each Fe is +3.

A	Confusion set	Reads the subscript 2 as the charge.
B	Dropped element	Leaves out the Roman numeral that iron needs.
C	Adjacent category	Prefixes are for molecular compounds.
D	Correct	Each iron is +3.

6. Answer B Concept 4

A teacher ~~who orders model kits every August~~ asks students to build ammonia, NH_3 . The central nitrogen has three bonds and one lone pair. ~~The kit uses blue balls for nitrogen.~~ What is the molecular geometry?

Target word: molecular shape **Decides:** Four electron domains, one of them a lone pair.

A	Adjacent category	Bent has two atoms and two lone pairs, like water.
B	Correct	Three atoms around a tetrahedral electron arrangement with one lone pair.
C	Dropped element	Ignores the lone pair.
D	Confusion set	That is the electron geometry, not the molecular shape.

7. Answer B Concept 5

Propane ~~from a 20-pound grill tank~~ burns: $\text{C}_3\text{H}_8 + __ \text{O}_2 \rightarrow 3\text{CO}_2 + 4\text{H}_2\text{O}$. ~~The grill was bought on sale.~~ What coefficient balances O_2 ?

Target word: O_2 coefficient **Decides:** Products have $3(2) + 4(1) = 10$ O atoms.

A	Confusion set	Adds product coefficients, counting molecules, not atoms.
B	Correct	$10 \text{ O atoms} / 2 \text{ per } \text{O}_2 = 5$.
C	Adjacent category	Changing a formula changes the substance; only coefficients change.
D	Dropped element	Counts O atoms but forgets O_2 carries two.

8. Answer B Concept 6

In a demonstration ~~that takes place behind a safety shield~~, 4.0 g of H_2 reacts with excess O_2 : $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$. ~~The room holds 24 students.~~ What mass of water forms? ($\text{H}_2 = 2.0 \text{ g/mol}$, $\text{H}_2\text{O} = 18.0 \text{ g/mol}$)

Target word: mass of product **Decides:** 2.0 mol H_2 gives 2.0 mol H_2O .

A	Dropped element	Uses 1 mol of water, skipping the 2:2 ratio.
B	Correct	$2.0 \text{ mol} \times 18.0 \text{ g/mol} = 36 \text{ g}$.
C	Inverted operation	Multiplies by 2 again instead of using the 1:1 ratio.
D	Wrong stem number	Assumes the mass of H_2 equals the mass of water.

9. Answer A Concept 6 • transfer

For $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$, how many grams of O_2 are needed to react completely with 4.0 g of H_2 ? ($\text{H}_2 = 2.0 \text{ g/mol}$, $\text{O}_2 = 32.0 \text{ g/mol}$)

Target word: solve for reactant mass **Decides:** 2.0 mol H_2 needs 1.0 mol O_2 .

A	Correct	$1.0 \text{ mol} \times 32.0 \text{ g/mol} = 32 \text{ g}$.
B	Inverted operation	Flips the 2:1 ratio.
C	Confusion set	Uses the molar mass of an O atom, not O_2 .
D	Wrong stem number	Assumes equal masses react.

10. Answer D Concept 7

Students heat ice from -20°C ~~using a hot plate with a digital display.~~ At 0°C the thermometer holds steady for several minutes while heating continues. ~~The lab has two windows.~~ What is the added energy doing during that time?

Target word: energy during the flat segment **Decides:** The flat section is melting.

A	Absolute language	Energy still enters; it changes state instead of temperature.
B	Confusion set	Temperature is truly constant during melting.
C	Adjacent category	Melting breaks forces between molecules, not bonds within them.
D	Correct	Energy goes to the phase change, not to raising temperature.

11. Answer A Concept 7 • transfer

Energy is added steadily to a pure substance, but its temperature does not change for a while. What is happening?

Target word: reverse from observation to cause **Decides:** Constant temperature under steady heating = phase change.

A	Correct	Phase changes absorb energy at constant temperature.
B	Confusion set	A zero specific heat would make temperature jump, not stall.
C	Adjacent category	A pure substance holding at one temperature is the pattern of a phase change.
D	Inverted operation	The stem says energy is being added.

12. Answer A Concept 8

A teacher preparing 1 M sulfuric acid ~~for a lab of 30 students~~ has a bottle of concentrated acid, ~~bought from a supplier the school has used for years,~~ and a flask of distilled water. What is the safe procedure?

Target word: safe dilution order **Decides:** Acid goes into water.

A	Correct	The water absorbs the heat released and prevents spattering.
B	Absolute language	Order matters for safety, not concentration.
C	Inverted operation	Water into concentrated acid can boil and spray acid.
D	Adjacent category	Handling concentrated acid is a teacher task; students get prepared dilute solutions.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	7	
Wrong stem number	A figure copied from the prompt that is not the answer.	5	
Adjacent category	The right idea in the wrong bucket.	7	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	6	
Absolute language	Always, never, guarantee — where judgment applies.	2	
Confusion set	Every option is real; the test is exact role-matching.	9	
Camouflage	Answers a stem fact that was never the question.	0	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Electron configuration	Fill orbitals in order (1s, 2s, 2p, 3s, 3p, 4s, 3d ...) until the electron count equals the atomic number of the neutral atom.	Adding or dropping electrons, or treating elements in the same group as the same element.	An atomic number or a configuration string.
2	Ionization energy trend	First ionization energy generally increases left to right across a period and bottom to top in a group.	Reversing the trend, or over-applying the small half-filled and filled subshell exceptions.	"Hardest to remove an electron" or "highest ionization energy."
3	Formulas from ion charges	Ionic formulas balance total positive and negative charge with the smallest whole-number ratio; Roman numerals give a transition metal's charge.	Writing one of each ion, or reading a subscript as the metal's charge.	A metal and nonmetal with different charges, or a transition metal name.
4	VSEPR molecular geometry	Electron domains set the electron geometry; lone pairs occupy space but the molecular shape names only atom positions.	Ignoring lone pairs, or naming the electron geometry as the molecular shape.	A central atom with one or more lone pairs.
5	Balancing equations	Change only coefficients, never subscripts, until each element has the same number of atoms on both sides.	Changing a subscript or counting molecules instead of atoms.	An unbalanced equation, often a combustion.
6	Mole-ratio stoichiometry	Convert grams to moles, use the balanced equation's mole ratio, then convert back to grams.	Going straight from grams to grams, or ignoring coefficients.	A mass of one substance and a question about another.
7	Heating curves and phase changes	During a phase change added energy breaks intermolecular attractions, so temperature stays constant.	Assuming added heat always raises temperature.	A flat section on a heating curve.
8	Lab safety: diluting acids	Always add concentrated acid slowly to water, not water to acid, with goggles, in a fume hood when needed.	Pouring water into concentrated acid, which can boil and spatter.	Preparing a dilute acid from a concentrated stock.

Course scope (public description): General Chemistry I with Lab for graduates provides an introduction to the field of chemistry to already-licensed teachers seeking an additional license or endorsement in secondary chemistry. This course provides students with opportunities to examine the electronic structure of atoms, study periodic trends, name chemical compounds, write chemical formulas, determine the structure of molecules, balance chemical reactions, and discover the changing states of matter. Laboratory experiences facilitate the study of matter and the application of laboratory safety and maintenance procedures. Concepts in Science is a prerequisite for this course.